

Lab-4 Assignment:

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Q1.Code :

```
main.cpp
1  #include <iostream>
2  using namespace std;
3
4  class HashTable {
5      int table[10];
6      int size;
7
8  public:
9      HashTable() {
10         size = 10;
11
12         for (int i = 0; i < size; i++) {
13             table[i] = -1;
14         }
15     }
16
17     void insert(int key) {
18         int index = key % size;
19
20         for (int i = 0; i < size; i++) {
21             int pos = (index + i) % size;
22
23             if (table[pos] == -1) {
24                 table[pos] = key;
25                 return;
26             }
27         }
28
29         cout << "Hash table is full.\n";
30     }
31 }
```

```

32 void display() {
33     cout << "\nHash Table:\n";
34
35     for (int i = 0; i < size; i++) {
36         cout << i << " -> ";
37
38         if (table[i] == -1)
39             cout << "Empty";
40         else
41             cout << table[i];
42
43         cout << endl;
44     }
45 }
46 };
47
48 int main() {
49     HashTable ht;
50
51     int n, key;
52
53     cout << "Enter number of elements: ";
54     cin >> n;
55
56     cout << "Enter elements:\n";
57     for (int i = 0; i < n; i++) {
58         cin >> key;
59         ht.insert(key);
60     }
61     ht.display();
62     return 0;
63 }

```

Q1.Output:

```

Output

Enter number of elements: 5
Enter elements:
10
20
25
35
15

Hash Table:
0 -> 10
1 -> 20
2 -> Empty
3 -> Empty
4 -> Empty
5 -> 25
6 -> 35
7 -> 15
8 -> Empty
9 -> Empty

```

Q2.Code :

main.cpp

```
1  #include <iostream>
2  using namespace std;
3
4  class HashTable {
5      int table[10];
6      int size;
7
8  public:
9      HashTable() {
10         size = 10;
11
12         for (int i = 0; i < size; i++) {
13             table[i] = -1;
14         }
15     }
16
17     void insert(int key) {
18         int index = key % size;
19
20         for (int i = 0; i < size; i++) {
21             int pos = (index + i * i) % size;
22
23             if (table[pos] == -1) {
24                 table[pos] = key;
25                 return;
26             }
27         }
28
29         cout << "Hash table is full.\n";
30     }
31
32     void display() {
33         cout << "\nHash Table:\n";
34
35         for (int i = 0; i < size; i++) {
36             cout << i << " -> ";
37
38             if (table[i] == -1)
39                 cout << "Empty";
40             else
41                 cout << table[i];
42
43             cout << endl;
44         }
45     }
46 };
47
```

Q2.Output:

```
Output

Enter number of elements: 5
Enter elements:
10
20
25
35
15

Hash Table:
0 -> 10
1 -> 20
2 -> Empty
3 -> Empty
4 -> Empty
5 -> 25
6 -> 35
7 -> Empty
8 -> Empty
9 -> 15

=== Code Execution Successful ===
```

Q3.Code:

```
main.cpp
1  #include <bit/stdc++.h>
2  using namespace std;
3
4  int main() {
5      int n, target;
6
7      cout << "Enter number of elements: ";
8      cin >> n;
9
10     int arr[n];
11
12     cout << "Enter elements:\n";
13     for (int i = 0; i < n; i++) {
14         cin >> arr[i];
15     }
16
17     cout << "Enter target value: ";
18     cin >> target;
19 }
```

```

20 unordered_set<int> seen;
21
22 for (int i = 0; i < n; i++) {
23     int required = target - arr[i];
24
25     if (seen.find(required) != seen.end()) {
26         cout << "Pair found: " << required << " + "
27         << arr[i] << " = " << target << endl;
28
29         return 0;
30     }
31
32     seen.insert(arr[i]);
33 }
34
35 cout << "No such pair exists." << endl;
36
37 return 0;
38 }

```

Q3.Output:

Output

Enter number of elements: 6

Enter elements:

2

27

11

10

15

12

Enter target value: 27

Pair found: 15 + 12 = 27

=== Code Execution Successful ===

Q4.Code:

```
main.cpp
1  #include <iostream>
2  #include <unordered_map>
3  using namespace std;
4
5  int main() {
6      int n;
7
8      cout << "Enter number of elements: ";
9      cin >> n;
10
11     int arr[n];
12
13     cout << "Enter elements:\n";
14     for (int i = 0; i < n; i++) {
15         cin >> arr[i];
16     }
17
18     unordered_map<int, int> frequency;
19
20     for (int i = 0; i < n; i++) {
21         frequency[arr[i]]++;
22     }
23
24     cout << "\nFrequency of elements:\n";
25
26     for (auto item : frequency) {
27         cout << item.first << " -> " << item.second << endl;
28     }
29
30     return 0;
31 }
```

Q4.Output:

```
Output

Enter number of elements: 5
Enter elements:
10
10
20
20
30

Frequency of elements:
30 -> 1
20 -> 2
10 -> 2

=== Code Execution Successful ===
```